

Lecture 2

1/23/26
JRA

Homework assignments, due dates, etc. will be posted on Moodle.

Previous lecture - course overview

Today's lecture - start w/ substance

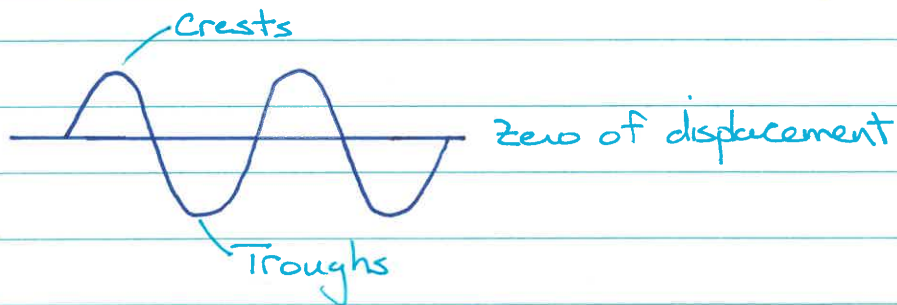
Waves are basis of much imaging

- Wave interacts w/ object
- Perturbations of wave by object used to form image

Attributes of Waves

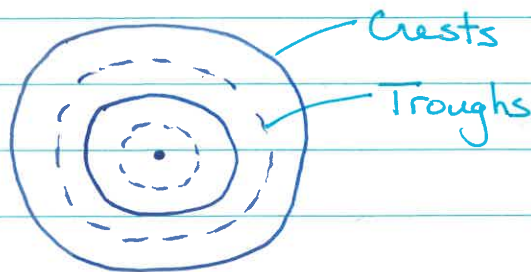
• 1D

Sinusoidal wave



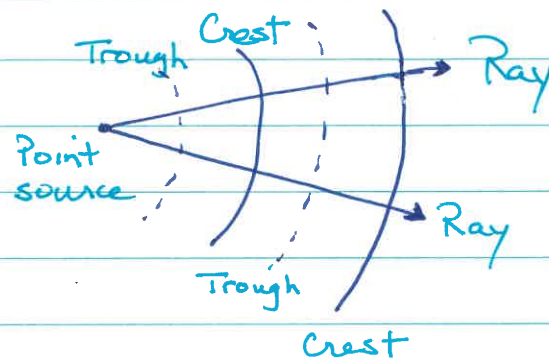
• 2D

Surface wave



Example is
pebble in water
causing circular waves

- 3D
Spherical waves



Rays point in direction of wave propagation

Start w/ point source, like a fluorophore

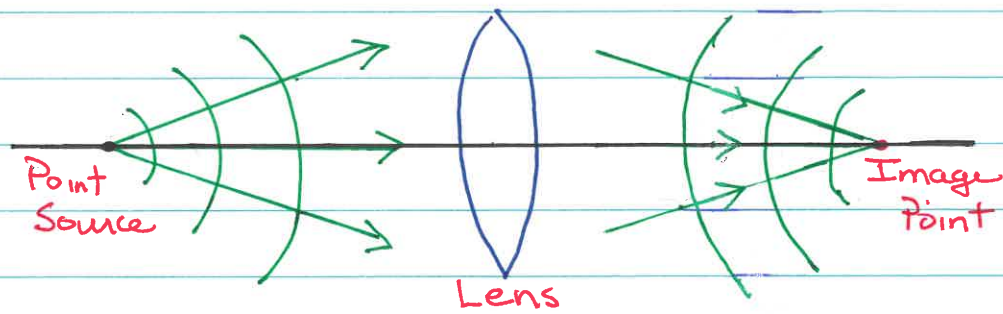
Point source emits spherical waves, which diverge (or expand) away from source

This class focuses on 3D waves

Define two pertinent terms from 3D wave diagram:

- Wavefronts - locations of crests & troughs
- Rays - straight lines $\left\{ \begin{array}{l} \text{Perpendicular } (\perp) \text{ to} \\ \text{wavefronts} \\ \text{Point in the direction of} \\ \text{energy propagation} \end{array} \right.$

Lenses, including the eye, can form an image of a source of spherical waves by converting the diverging spherical waves coming from the source to converging spherical waves



The whole game w/ vision & imaging is to have a device that causes waves to converge.

Do this w/ a lens (as shown above)

How do we see things?

Object self-luminous, like the sun or a fluorophore that emits light

Object reflects or scatters light incident on it
(See cat picture in PowerPoint slides)

Need lens to reconfigure light to form image

ANIMATION

Physlet 35.2 Ray Diagrams

Looked at how lenses form images

Different types of lenses

Converging

Diverging

Used to correct farsightedness

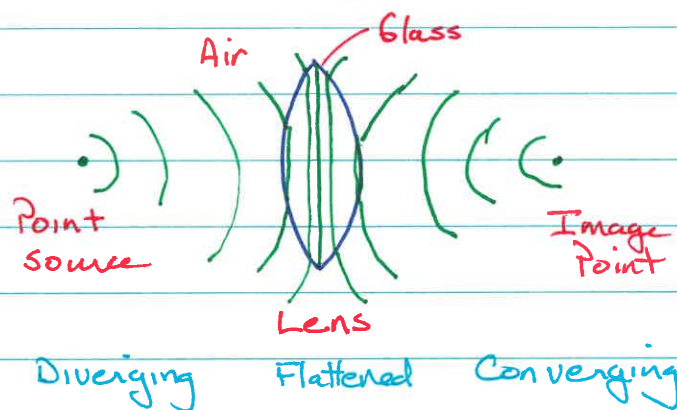
Used to correct nearsightedness

Different placements of lens relative to object & image

This class will focus mostly on converging lenses

There are a couple of ways to understand how converging lenses reshape diverging waves

- Wave Speed Argument



Converging lens
is wider in middle
than at edges

Light slows in denser media, like glass

Light hits bulging center of lens first

Center of wave slows, because wave slower in glass, while edges move faster & catch up $\Rightarrow$ wave flattens

This requires a precise lens shape

On exit from lens, edges are released first

The net effect is to convert the incoming diverging wave into an outgoing converging wave.

The speed of light in air & glass can be calculated from the following formula:

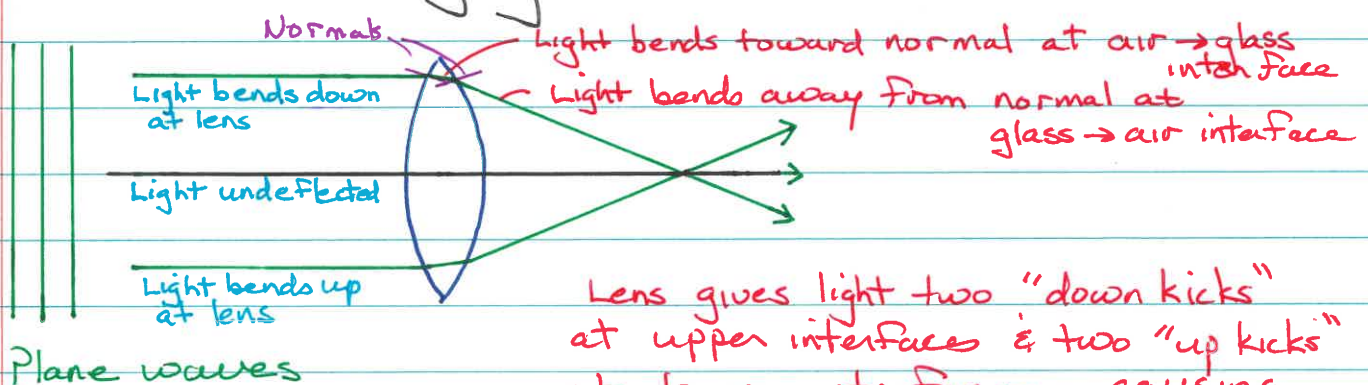
$$v = \frac{c}{n}$$

c = speed of light in vacuum = $3 \cdot 10^8 \text{ m/s}$
 n = refractive index

$$n = \begin{cases} 1 & \text{air} \\ 1.5 & \text{glass} \end{cases}$$

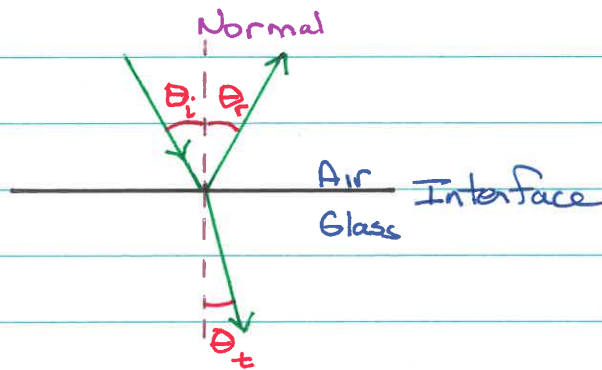
• Ray Bending Argument

Ray bending at the air/glass (lens) interface is another way to understand how a converging lens works to convert a diverging wave into a converging wave



Lens gives light two "down kicks" at upper interfaces & two "up kicks" at lower interfaces, causing light on top & bottom to bend toward middle (i.e., converge)

Waves are reflected and refracted at an interface



Normal = perpendicular to interface

i = incident

r = reflected

t = transmitted

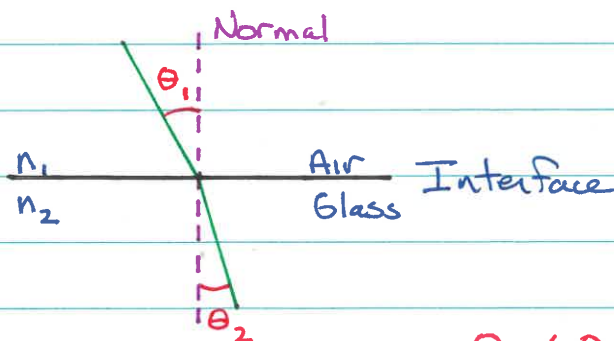
Law of Reflection

$$\theta_r = \theta_i$$

Reflection angle = incidence angle at a planar interface

Law of Refraction (Snell's Law)

Light bends toward normal when going from air to glass & away from normal when going from glass to air



$$\begin{aligned} n_1 &= n_{\text{air}} = 1.00 \\ n_2 &= n_{\text{glass}} = 1.52 \end{aligned}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\theta_2 < \theta_1 \text{ if } n_2 > n_1$$

$$\theta_2 > \theta_1 \text{ if } n_2 < n_1$$

Summary - use a converging (or bulging) lens to bring light to focus & form an image

Lenses can be used to form images of different sizes, relative to the original object, depending on application

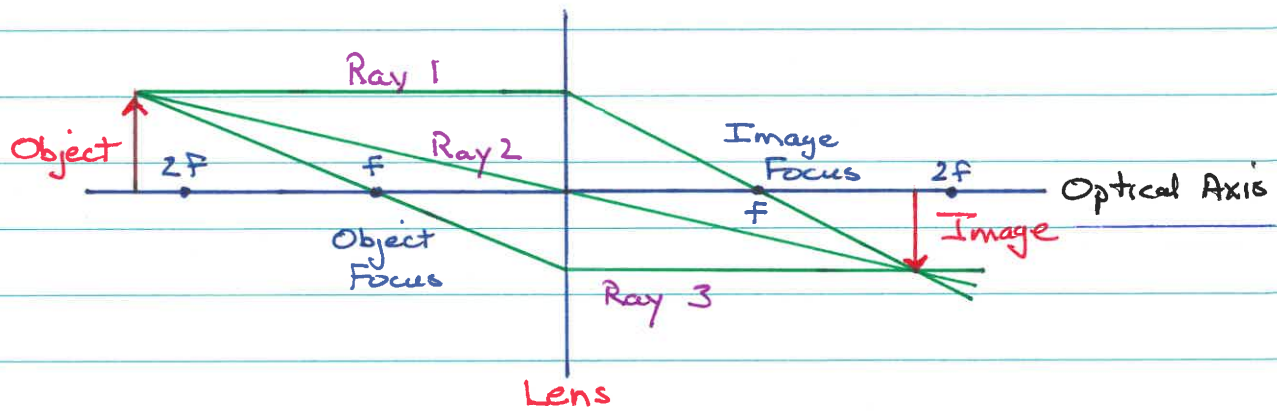
- Eye & Camera $\Rightarrow$ Minify
- Photocopier $\Rightarrow$ No change
- Microscope $\Rightarrow$ Magnify

Lens performance (image size, image location) can be assessed w/ two tools

- Ray Tracing (graphical)
- Lens Equations (mathematical)

Look at both of these tools

Microscopes use complicated lens systems w/ many lenses. Start discussion w/ one "equivalent" lens.



Lens - "thin lens"; represent by a vertical line ($|$), where all bending takes place on the line

Object - an arrow ($\uparrow$)

Focal Points - lens has two symmetric "Focal points", defined by where lens brings parallel incident rays to focus; labeled w/ "F"

Object (arrow) will emit rays in many directions.

Lens will only capture a subset of emitted rays

Fact that lens only captures some rays is what limits resolution.

Look at 3 key rays to determine size & location of image

- Ray 1 - parallel to optical axis on object side, then thru image focus
- Ray 2 - thru center of lens, undeflected
- Ray 3 - thru object focus, then parallel to optical axis on image side

ANIMATION

Physlet 35.2

Ray Diagrams

Looked at ray traces, image sizes, & image positions for a variety of object positions (measured relative to f)

Object position $\begin{cases} > 2f & \text{Minified inverted image} \\ = 2f & \text{Same-sized inverted image} \\ > f \text{ AND } < 2f & \text{Magnified inverted image} \end{cases}$

These three scenarios all created "real" images in which the rays (& the light) converged

To get magnification, object must be close to object focal point, & so close to lens

Simple microscope uses two lenses, an "objective" & an "eyepiece", to increase magnification

• Objective	60-100x	} Total magnification
• Eyepiece	10x	
		= 600-1000x